X(3872) Analysis Update

From ψ(2S) reweighting to the current PbPb strategy

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2026.8.5



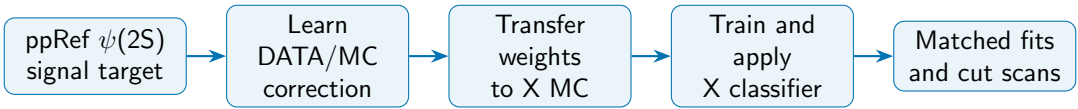








Recap: where the previous update stopped



ppRef: end-to-end prototype completed

- Signal extraction, reweighting, X MC application
- Weighted/unweighted ML, matched fits, cut scans
- Technical workflow established; physics choice not frozen

PbPb: no reliable $\psi(2S)$ target

- Cumulative selections and S2/S3 fits checked
- No sufficiently robust peak for a signed-sWeight target
- PbPb-specific reweighting could not start from this route

This update revisits the ppRef assumptions before choosing a PbPb strategy.

Comments postponed

Comment	Reason for postponement
Use ML AUC to decide whether more inputs are better?	MC-based AUC is not a reliable standalone criterion for variable inclusion.
Add `B_alpha`?	Not a current bottleneck; no new training or validation in this update.
Add muon variables?	Revisit only after the reweighting and transfer strategy is defined.
Move `BQvalue` into the fiducial region?	A broad bias question requiring a dedicated selection and systematic study.

What AUC can tell us

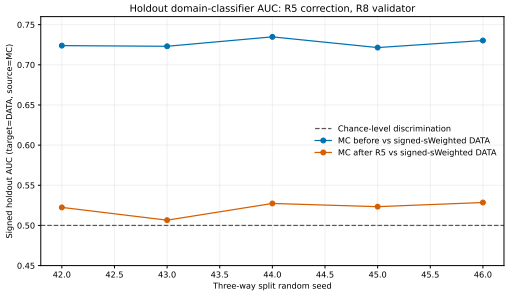
Separation power on the simulated validation sample.

What it cannot establish

DATA modelling, transfer closure, or final analysis robustness.

Feature expansion is paused until the correction strategy and validation scope are fixed.

What does AUC = 0.5 mean here?



This diagnostic

- Classifies **DATA target vs. MC domain**
- Positive: signed-sWeighted $\psi(2S)$ DATA
- Negative: $\psi(2S)$ MC before/after R5 correction

Different from ML AUC

Signal-vs-background AUC should be high; domain AUC should approach 0.5 after correction.

Five holdout splits after reweighting:

range 0.5066–0.5285; mean 0.5217.

From the KS test to a signed-weight KS-like distance

Standard two-sample KS test

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)| .$$

- Unweighted, monotonic empirical CDFs
- A KS p-value follows under standard assumptions

Our signed-weight distance

$$D = \sup_x \left| F_{\mathrm{MC}}(x) - F^{\mathrm{signed}}_{\mathrm{target}}(x) \right| .$$

- Negative sWeights are retained
- Signed CDF may decrease or leave $[0, 1]$
- Descriptive effect size, not a KS p-value

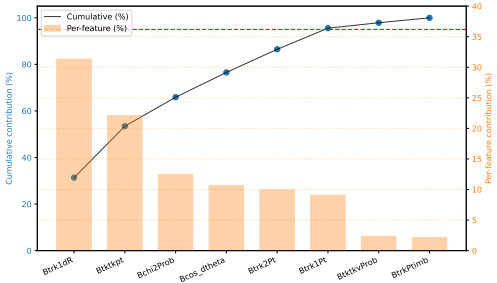
Completed bootstrap study: 500 paired replicas; resample DATA/MC, retain signed event weights, retrain each reweighter, and evaluate an independent holdout.

Covered	Not covered
Finite-event fluctuations; retraining	Fit model/range; sWeight construction

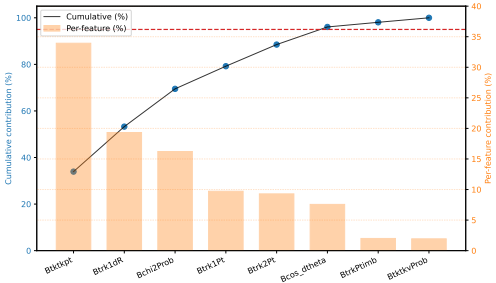
Later recipe scans use point estimates only; the bootstrap is not repeated by default.

SHAP importance in the new nominal workflow

R6range2-weighted classifier



Unweighted classifier



R6range2 correction used by this artifact: $\texttt{Bcos_dtheta} \in [-1, 1]$, $\texttt{Btktkpt} \in [2, 10]$, $\texttt{Bchi2Prob} \in [0, 1]$, $\texttt{Btrk2Pt} \in [0.5, 4.5]$, $\texttt{Btrk1Pt} \in [0.5, 4.5]$, $\texttt{Btrk1dR} \in [0, 0.5]$.

New nominal chain: ppRef X fit $\rightarrow$ reweighting $\rightarrow$ PbPb weighted-vs-unweighted ML comparison.

SHAP importance versus ppRef X closure

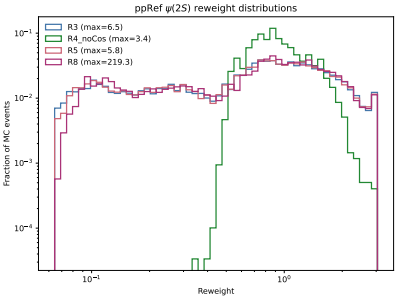
Variable	SHAP fraction: unweighted → weighted	KS-like: before → after	Pattern
`Btrk1dR`	19.33% → 31.34%	0.051 → 0.039	importance rises
`Btktkpt`	33.94% → 22.10%	0.242 → 0.037	both decrease
`Bchi2Prob`	16.22% → 12.47%	0.123 → 0.032	both decrease
`Bcos_dtheta`	7.57% → 10.63%	0.260 → 0.040	importance rises
`Btrk2Pt`	9.29% → 9.94%	0.093 → 0.017	nearly stable
`Btrk1Pt`	9.72% → 9.06%	0.176 → 0.043	mild decrease

Corrected variables remain dominant, but no simple monotonic SHAP–closure relation is observed.

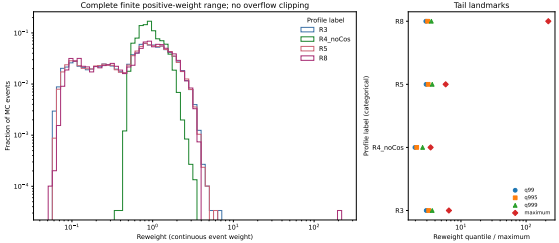
SHAP: PbPb classifiers; KS-like: ppRef X R6range2 closure. The comparison is qualitative.

Why did the previous plot not show the maximum weight?

Previous update



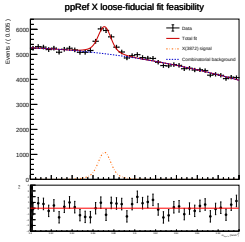
Corrected full-range view



- The previous histogram stopped at the **99.5% positive-weight quantile**; its legend nevertheless reported the true maximum.
- The corrected view spans the complete finite positive-weight range: the long tail is visible and the R8 maximum = 219.262 is real.

How is the ppRef X validation target constructed?

Comment from the previous update: compare the X sPlot target with $\psi(2S)$-reweighted X MC and quantify the agreement with a KS-like distance.



Technically accepted fit: status $= 0$, covQual $= 3$; $N_X = 3795 \pm 288$.

Independent X target

- $7.5 < p_T(B) < 50$, $|y| < 2.4$, `BQvalue`< 0.15
- Yield-only sPlot refit; signed weights retained
- $N_{\text{eff}} = 393.74$ from 194,886 candidates

Transfer test

signed-sWeighted ppRef $\psi(2S)$ DATA + $\psi(2S)$ MC

↓

learn five-variable correction

↓

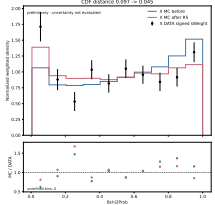
apply correction to ppRef X MC

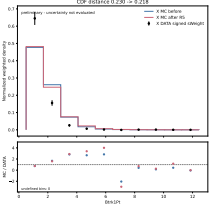
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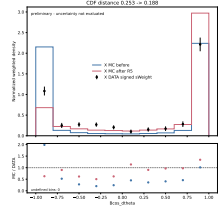
compare with X signed-sPlot target

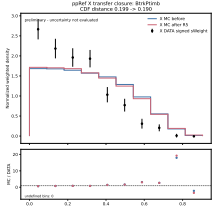
Inputs: `Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk2Pt`, `Btrk1Pt`; correction selection: $p_T(B) > 7.5$, $|y| < 2.4$, `BQvalue`< 0.15.

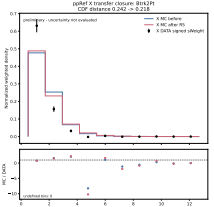
Most variables move closer to the X signed-sPlot target

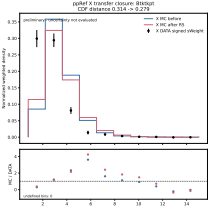






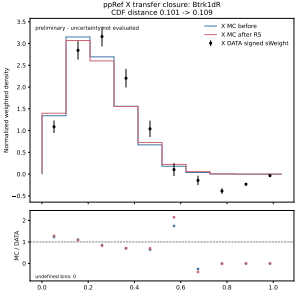






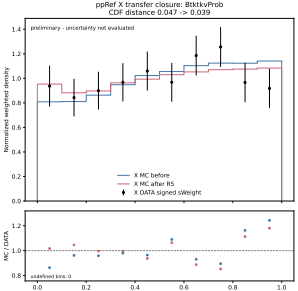
Six variables show point-estimate improvement after transferring the $\psi(2S)$-derived weights to X MC.

The remaining variables limit the transfer reliability



`Btrk1dR`: **worsens**

$0.1012 \rightarrow 0.1091$



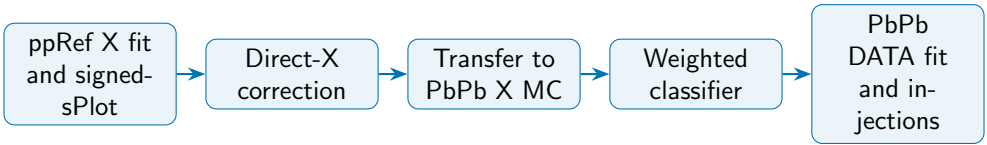
`BtktkvProb`: **small improvement**

$0.0472 \rightarrow 0.0393$

Overall diagnostic	Before $\rightarrow$ after
Eight-variable mean KS-like distance	$0.1855 \rightarrow 0.1609$
Variables improved	$7/8$

The tested $\psi(2S)$-derived weights do not provide a reliable description of X across all validation variables.

A new X-based nominal candidate



Why change the route?

- ψ(2S)-derived correction did not reliably describe X
- PbPb ψ(2S) did not provide a usable sPlot target
- Use the accepted ppRef X target directly

The correction is learned from the ppRef X signed-sPlot target and consumed only within a controlled common support.

Unrestricted X-target reweighting is unstable

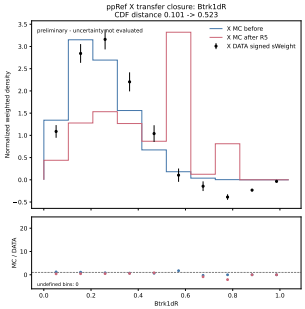
Unrestricted five-variable correction

- `Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk2Pt`, `Btrk1Pt`
- Fiducial: $7.5 < p_T(B) < 50$, $|y| < 2.4$, `BQvalue`< 0.15; no controlled input-variable ranges

Assessment of unrestricted X reweighting

Maximum event weight	23910.72
Weighted-MC N_{eff}	11.79
$\overline{D}$	$0.1855 \rightarrow 0.3315$
D(`Btrk1dR`)	$0.1012 \rightarrow 0.5227$

Effective sample size: $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, the equivalent number of unit-weight events after reweighting.



Unrestricted correction strongly distorts `Btrk1dR`.

The unrestricted correction cannot be transferred; common phase-space control is required.

Correction variable sets

Common fiducial: $7.5 < p_T(B) < 50$, $|y| < 2.4$, `BQvalue`< 0.15.

Variable set	Reweighting variables
R5	`Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk2Pt`, `Btrk1Pt`
R6	R5 + `Btrk1dR`
R5v2	`Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk1Pt`, `Btrk1dR`
R6v2	R5v2 + `Btrk2dR`

R5 → R6

Add `Btrk1dR` after its closure worsens with R5range.

R5v2 → R6v2

Test track-1-only inputs, then add `Btrk2dR` explicitly.

Variable-set changes and phase-space range changes are tested separately.

Controlled ranges and tested variants

Ranges shared whenever the variable is present

- `Bcos_dtheta` $\in [-1, 1]$
- `Bchi2Prob` $\in [0, 1]$
- `Btrk1Pt`, `Btrk2Pt` $\in [0.5, 4.5]$
- `Btrk1dR`, `Btrk2dR` $\in [0, 0.5]$

Only `Btktkpt` changes between range variants

- R5range, R6range: $[1, 10]$
- R6range2, R5v2range, R6v2range: $[2, 10]$
- R5v2range2, R6v2range2: $[2, 8]$

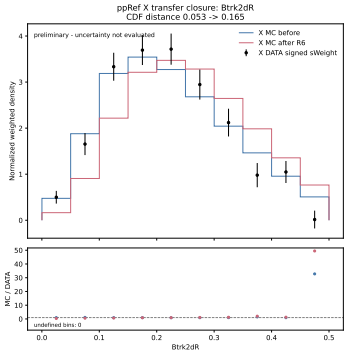
Representative outcomes

Configuration	max w	N_{eff}	$\overline{D}$ after
R5range	74.11	17106	0.0581
R6range	951.65	2085	0.0376
R6range2	16.45	34641	0.0350
R5v2range	15.84	32920	0.0354
R6v2range	38.11	28169	0.0724
R5v2range2	17.53	33836	0.0374
R6v2range2	42.33	25687	0.0738

Both track-dR inputs in R6v2range2

$D(\texttt{Btrk1dR}) = 0.03552 \rightarrow 0.08319$
$D(\texttt{Btrk2dR}) = 0.04968 \rightarrow 0.07047$

The second-track angular distance remains mismodelled



R6range2, evaluated within $0 < \texttt{Btrk2dR} < 0.5$

Independent closure check

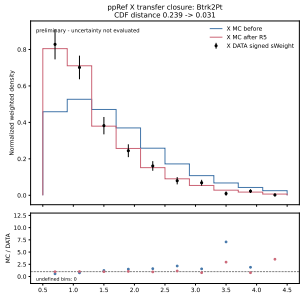
Configuration	D before	D after
R5range	0.1051	0.2014
R6range2	0.0525	0.1651
R5v2range	0.0493	0.1746
R6v2range	0.0493	0.0840

- `Btrk2dR` is excluded from the eight-variable mean
- Adding one or both track-dR inputs does not restore closure
- Every tested configuration gives a larger $D(\texttt{Btrk2dR})$

The aggregate improvement does not extend to the independent `Btrk2dR` check.

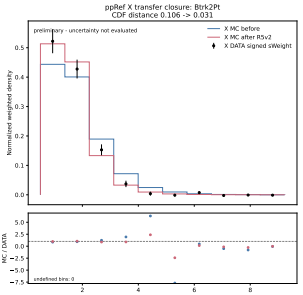
Momentum closure improves without resolving track-dR

R5range: includes `Btrk2Pt`



$D(\texttt{Btrk2Pt}):\ 0.239176 \rightarrow 0.031423$

R5v2range: excludes `Btrk2Pt`

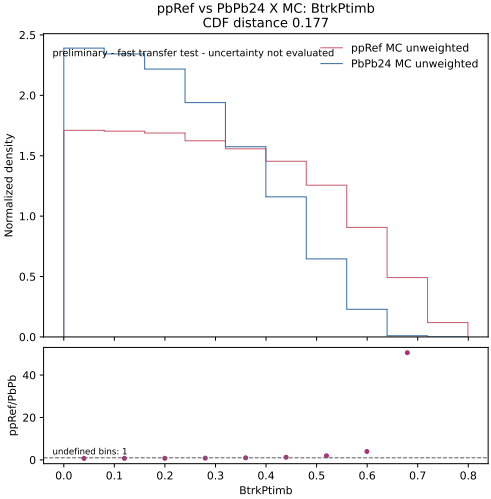


$D(\texttt{Btrk2Pt}):\ 0.106097 \rightarrow 0.031412$

Contrast: `Btrk2Pt` improves strongly whether it is a correction input or an independent check, while `Btrk2dR` worsens in all tested configurations.

R6range2 is a pragmatic stability–closure compromise; no tested variant resolves `Btrk2dR`.

Transfer is restricted to a common phase space



1. Transfer domain

Learn inside ppRef R6range2 support. Apply with tighter PbPb: $10 < \texttt{Bpt} < 50$, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$.

2. MC-domain baseline

Compare unweighted ppRef/PbPb MC. Adding full support, including $\texttt{Btktkpt} \in [2, 10]$, changes mean $\overline{D}$ from 0.0943 to 0.0843.

3. Apply contract

Reweight 32,160 in-support events; keep 22,908 outside at unity and exclude them downstream.

Weighted and unweighted classifiers define the fit inputs

Shared training definition

- Background: PbPb DATA, `Bmass`$\in [3.75, 3.80] \cup [3.95, 4.00]$ GeV
- Candidate-level 75/25 split; common support

Only the signal model changes

- Weighted: transferred correction on PbPb X MC
- Unweighted: unit-weight PbPb X MC
- Same fit workflow consumes both scored samples

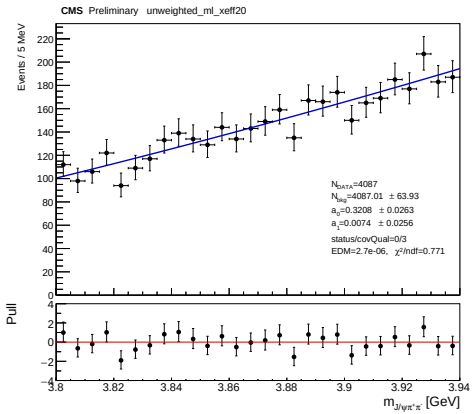
Eight shared inputs: `Bchi2Prob`, `Btrk1dR`, `BtrkPtimb`, `Btrk1Pt`, `Btrk2Pt`, `BtktkvProb`, `Bcos_dtheta`, `Btktkpt`.

Classifier	signal weighting	train AUC	test AUC
Weighted signal	transferred event weights	0.8777	0.8391
Unit signal	unit event weights	0.9115	0.8958

Interpretation boundary: the classifiers see different signal distributions; a higher MC AUC may absorb DATA–MC mismatch and does not select the physical model.

AUC and SHAP describe model response; the decision requires observed spectra and injection tests.

Observed data after the unweighted classifier



Selected spectrum

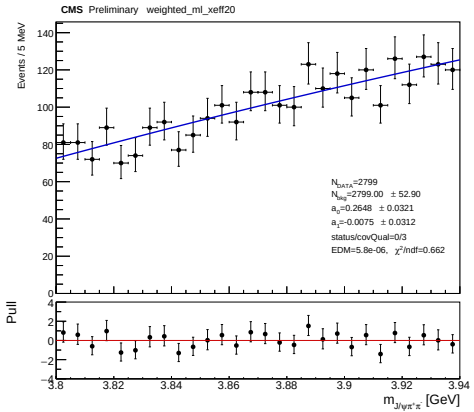
- 20% weighted-reference X efficiency
- 4087 DATA candidates
- `Bmass`: 3.80–3.94 GeV
- Smooth second-order background

Fit quality

- status $= 0$
- covQual $= 3$
- $\chi^2/\mathrm{ndf} = 0.771$

No narrow X peak is clearly visible; a smooth background-only model adequately describes this selected spectrum.

Observed data after the weighted classifier



Selected spectrum

- 20% weighted-reference X efficiency
- 2799 DATA candidates
- `Bmass`: 3.80–3.94 GeV
- Smooth second-order background

Fit quality

- status $= 0$
- covQual $= 3$
- $\chi^2/\mathrm{ndf} = 0.662$

The weighted classifier also yields no clearly visible narrow X peak in the observed spectrum.

The absence of a visible peak is not a transfer test by itself

	Weighted-reference X efficiency			
Classifier	20%	30%	40%	50%
Unweighted: DATA candidates	4087	10765	23422	43273
Unweighted: χ^2/ndf	0.771	0.979	0.981	0.762
Weighted: DATA candidates	2799	7742	16983	31399
Weighted: χ^2/ndf	0.662	1.217	0.917	0.569

- All eight background-only fits converge
- No clearly visible narrow X peak
- Smooth background gives an acceptable description

A leading explanation for the absent peak is residual mismatch between the reweighted MC and the signal-variable distributions.

A matched-signal injection tests the current workflow

Validation question

1. Generate pseudo-data with signal-variable distributions matching the current X MC.
2. Apply the current classifier selection and `Bmass` fit workflow.
3. Test whether the injected narrow signal peak and yield are recovered.

Signal-yield estimate

$$N_X^{\mathrm{after}} = N_\psi^{\mathrm{raw}} \times 1.08 \times \frac{\epsilon_X^{\mathrm{score}}}{\epsilon_\psi^{\mathrm{score}}}.$$

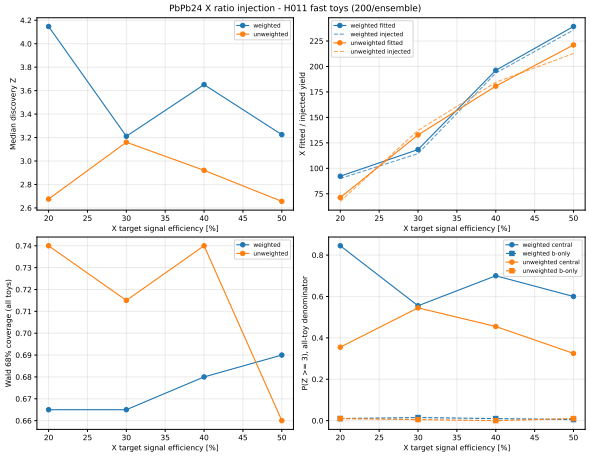
- N_ψ^{raw} from eight accepted PbPb DATA fits
- Expected yield evaluated after each score cut

Expected signal yields

X efficiency	weighted	unweighted
20%	90.0 ± 22.2	67.6 ± 17.7
30%	114.4 ± 32.3	137.2 ± 27.9
40%	193.5 ± 73.2	184.5 ± 40.8
50%	235.8 ± 63.2	212.7 ± 55.5

Fast-test assumptions: pre-score acceptance–efficiency ratio = 1; prompt-fraction ratio = 1; only the ψ(2S) fit statistical uncertainty is propagated.

Matched-model injections recover the assumed signal



Fast-test design

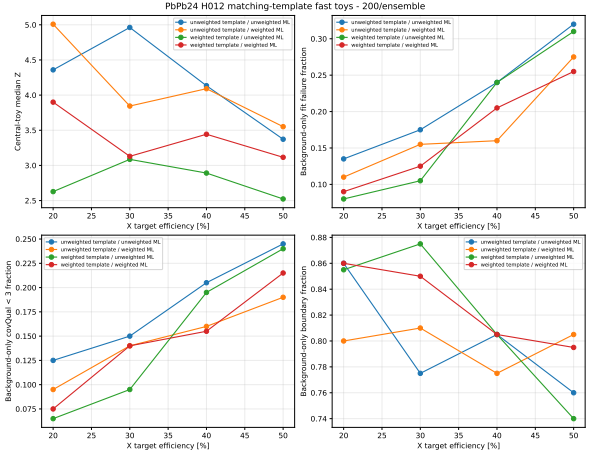
- Asimov fits at the central yield and $\pm 1\sigma_\psi$
- 200 toys with the central signal injection
- 200 background-only toys
- Generation and fit use matching signal models

Observed in toys

- Median expected Z: 2.66–4.15
- Central Asimov bias: -0.8% to -3.3%
- Background-only: 12–43% failure; 71.5–88% boundary

Matched-signal sensitivity does not validate the ppRef-to-PbPb transfer in data.

Template and classifier choices do not yet define an optimum



Matching-template matrix

- Each template × classifier: central and $\pm 1\sigma_{\psi}$ Asimov fits
- 200 central-injection + 200 background-only toys
- Median Z: 2.52–5.01

Remaining limitations

- Mean failure change: -0.0275
- No systematic covQual/boundary gain
- 7 central recovery warnings

Expected sensitivity depends jointly on template, classifier, and injected yield; no final model or working point is established.

Summary and Outlook

Question	Established	Remaining limitation
Extreme weights	Range control restores N_{eff}	
Variables/ranges	A working configuration	Not optimal
`Btrk1dR`/`Btrk2dR`		Failure persists across variants
Region transfer	Explicit extrapolation is excluded	No PbPb DATA closure
Observed spectra		No clearly visible narrow X peak
Injection checks	ML is working	

Possible ways forward

1. Test ppRef $\psi(2S)$ correction transfer within common support.
2. Compare variables/ranges with common validation gates.
3. Explore additional potentially useful input variables.

Thanks for listening!

Reporter: Lekai Yao

2026.8.5